

WAVES

Important formulas and concepts directly from NCERT-

❖ Wavelength(λ) and angular wave no.(k) –

$$\lambda = 2\pi/k$$

❖ Period(T) , angular frequency(ω) and frequency(ν)-

$$\omega = 2\pi/T$$

$$\nu = 1/T = \omega/2\pi$$

❖ Speed of a travelling wave-

$$v = \omega/k = \lambda/T$$

❖ Speed of transverse wave on stretched string-

$$v = \sqrt{T/\mu}$$

wavelength,

$$\lambda = v/\nu = \text{speed of wave in medium/frequency}$$

❖ Speed of longitudinal wave(sound wave)-

○ For gases or fluids,

$$v = \sqrt{B/\rho}$$

(B = bulk modulus)

○ For solids,

$$v = \sqrt{Y/\rho}$$

(Y = young's modulus)

➤ According to newton,

$$v = \sqrt{P/\rho}$$

Newton considered it isothermal process.

➤ Laplace correction – it is an adiabatic process

$$v = \sqrt{\gamma P/\rho}$$

❖ **Organ pipes-**

- **Closed organ pipe-** pipe closed at both ends.

Frequencies,

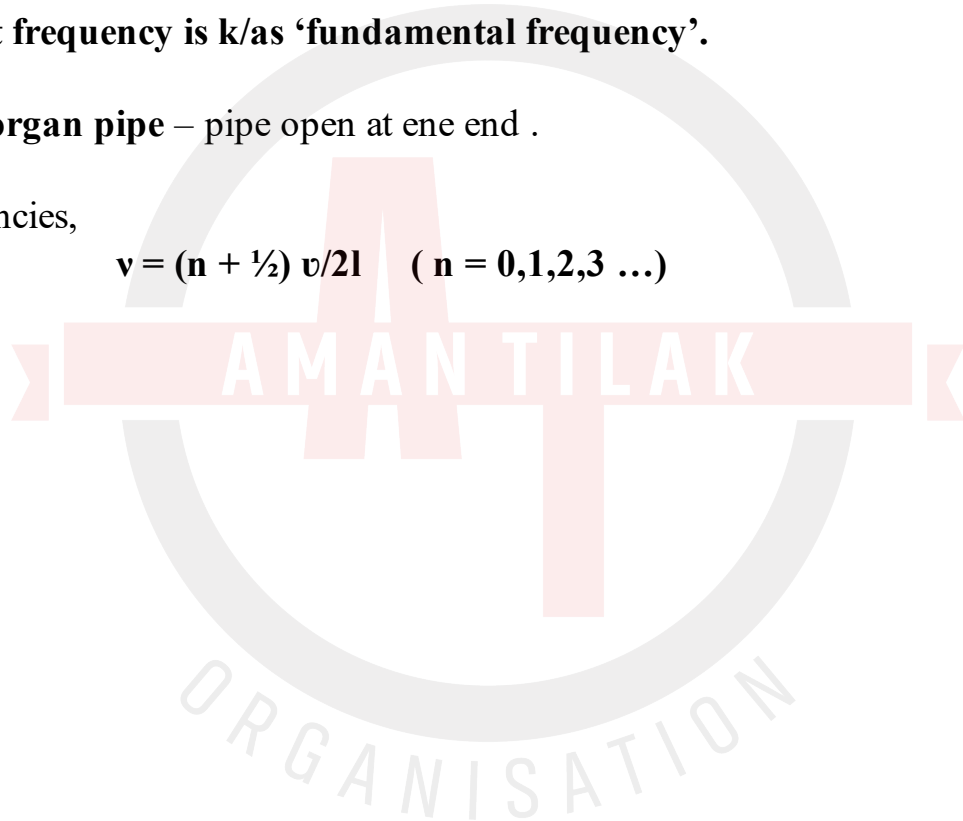
$$v = nv/2l \quad (n = 1, 2, 3 \dots)$$

Lowest frequency is k/as ‘fundamental frequency’.

- **Open organ pipe** – pipe open at one end .

Frequencies,

$$v = (n + \frac{1}{2}) v/2l \quad (n = 0, 1, 2, 3 \dots)$$



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SUMMARY

1. *Mechanical waves* can exist in material media and are governed by Newton's Laws.
2. *Transverse waves* are waves in which the particles of the medium oscillate perpendicular to the direction of wave propagation.
3. *Longitudinal waves* are waves in which the particles of the medium oscillate along the direction of wave propagation.
4. *Progressive wave* is a wave that moves from one point of medium to another.
5. *The displacement* in a sinusoidal wave propagating in the positive x direction is given by

$$y(x, t) = a \sin(kx - \omega t + \phi)$$

where *a* is the amplitude of the wave, *k* is the angular wave number, *ω* is the angular frequency, *$(kx - \omega t + \phi)$* is the phase, and *ϕ* is the phase constant or phase angle.

6. *Wavelength λ* of a progressive wave is the distance between two consecutive points of the same phase at a given time. In a stationary wave, it is twice the distance between two consecutive nodes or antinodes.
7. *Period T* of oscillation of a wave is defined as the time any element of the medium takes to move through one complete oscillation. It is related to the *angular frequency ω* through the relation

$$T = \frac{2\pi}{\omega}$$

8. Frequency ν of a wave is defined as $1/T$ and is related to angular frequency by

$$\nu = \frac{\omega}{2\pi}$$

9. Speed of a progressive wave is given by $v = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda\nu$

10. The speed of a transverse wave on a stretched string is set by the properties of the string. The speed on a string with tension T and linear mass density μ is

$$v = \sqrt{\frac{T}{\mu}}$$

11. Sound waves are longitudinal mechanical waves that can travel through solids, liquids, or gases. The speed v of sound wave in a fluid having bulk modulus B and density ρ is

$$v = \sqrt{\frac{B}{\rho}}$$

The speed of longitudinal waves in a metallic bar is

$$v = \sqrt{\frac{Y}{\rho}}$$

For gases, since $B = \gamma P$, the speed of sound is

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

12. When two or more waves traverse simultaneously in the same medium, the displacement of any element of the medium is the algebraic sum of the displacements due to each wave. This is known as the *principle of superposition* of waves

$$y = \sum_{i=1}^n f_i(x - vt)$$

13. Two sinusoidal waves on the same string exhibit *interference*, adding or cancelling according to the principle of superposition. If the two are travelling in the same direction and have the same amplitude a and frequency but differ in phase by a *phase constant* ϕ , the result is a single wave with the same frequency ω :

$$y(x, t) = \left[2a \cos \frac{1}{2} \phi \right] \sin \left(kx - \omega t + \frac{1}{2} \phi \right)$$

If $\phi = 0$ or an integral multiple of 2π , the waves are exactly in phase and the interference is constructive; if $\phi = \pi$, they are exactly out of phase and the interference is destructive.

14. A travelling wave, at a rigid boundary or a closed end, is reflected with a phase reversal but the reflection at an open boundary takes place without any phase change.

For an incident wave

$$y_i(x, t) = a \sin(kx - \omega t)$$

the reflected wave at a rigid boundary is

$$y_r(x, t) = -a \sin(kx + \omega t)$$

For reflection at an open boundary

$$y_r(x, t) = a \sin(kx + \omega t)$$

15. The interference of two identical waves moving in opposite directions produces *standing waves*. For a string with fixed ends, the standing wave is given by

$$y(x, t) = [2a \sin kx] \cos \omega t$$

Standing waves are characterised by fixed locations of zero displacement called *nodes* and fixed locations of maximum displacements called *antinodes*. The separation between two consecutive nodes or antinodes is $\lambda/2$.

A stretched string of length L fixed at both the ends vibrates with frequencies given by

$$\nu = \frac{n v}{2 L}, \quad n = 1, 2, 3, \dots$$

The set of frequencies given by the above relation are called the *normal modes* of oscillation of the system. The oscillation mode with lowest frequency is called the *fundamental mode* or the *first harmonic*. The *second harmonic* is the oscillation mode with $n = 2$ and so on.

A pipe of length L with one end closed and other end open (such as air columns) vibrates with frequencies given by

$$\nu = (n + \frac{1}{2}) \frac{v}{2L}, \quad n = 0, 1, 2, 3, \dots$$

The set of frequencies represented by the above relation are the *normal modes* of oscillation of such a system. The lowest frequency given by $v/4L$ is the fundamental mode or the first harmonic.

16. A string of length L fixed at both ends or an air column closed at one end and open at the other end or open at both the ends, vibrates with certain frequencies called their normal modes. Each of these frequencies is a *resonant frequency* of the system.
17. *Beats* arise when two waves having slightly different frequencies, ν_1 and ν_2 and comparable amplitudes, are superposed. The beat frequency is

$$\nu_{beat} = \nu_1 \sim \nu_2$$

POINTS TO PONDER

1. A wave is not motion of matter as a whole in a medium. A wind is different from the sound wave in air. The former involves motion of air from one place to the other. The latter involves compressions and rarefactions of layers of air.
2. In a wave, energy and *not the matter* is transferred from one point to the other.
3. In a mechanical wave, energy transfer takes place because of the coupling through elastic forces between neighbouring oscillating parts of the medium.
4. Transverse waves can propagate only in medium with shear modulus of elasticity. Longitudinal waves need bulk modulus of elasticity and are therefore, possible in all media, solids, liquids and gases.
5. In a harmonic progressive wave of a given frequency, all particles have the same amplitude but different phases at a given instant of time. In a stationary wave, all particles between two nodes have the same phase at a given instant but have different amplitudes.
6. Relative to an observer at rest in a medium the speed of a mechanical wave in that medium (v) depends only on elastic and other properties (such as mass density) of the medium. It does not depend on the velocity of the source.